EcoVault Face Verification for Field Research Data Security

Introduction

This project looks at whether face verification can realistically work as a second layer of security for EcoVault, a mobile app designed to protect sensitive field research data like species locations, survey notes, and environmental records. The idea is that after a user enters their PIN, the system uses a standard RGB image to check whether the face matches the enrolled user. Since this is a one-to-one verification problem rather than general face recognition, the focus is on metrics like false accept rate, false reject rate, and threshold performance instead of overall classification accuracy.

Data & Methodology

I used the Labeled Faces in the Wild dataset, which contains 5,985 images across 423 identities at a resolution of 62 by 47 pixels. I selected George W. Bush as the enrolled user because he had 530 available images, which gave the positive class much more variation compared to other identities with fewer samples. The setup treated George W. Bush as the positive class and all other identities as negatives, resulting in 530 positive images and 5,455 negative images.

The positives were split 60 percent for training, 20 percent for validation, and 20 percent for testing. For the negatives, I split at the identity level so that completely unseen impostor identities only appeared in the test set. This produced 3,526 training images, 1,176 validation images, and 1,283 test images, with the positive class making up about 9 percent of each split. To evaluate robustness, I created brightness and contrast labels using mean pixel values and standard deviation, then split the data into bright and dark groups using a median threshold.

I tested four different approaches in this setup. These included frozen VGG16 embeddings with cosine similarity, a small CNN trained from scratch, a fine-tuned VGG16 model, and a fine-tuned MobileNetV2 model that was designed to be more efficient for deployment. I evaluated performance using AUC, Equal Error Rate, and True Positive Rate at a False Accept Rate less than or equal to 0.5 percent, since those metrics better reflect real verification performance than standard accuracy.

Results & Metrics vs. Spec

The results show a clear gap between the baseline approaches and the fine-tuned models. Both the frozen VGG16 and the small CNN struggled to separate the enrolled user from impostors, while the fine-tuned models were able to learn much stronger identity-specific features and significantly improve performance.

Looking at the table, fine-tuned VGG16 is the strongest in terms of verification quality, with high AUC, lower EER, and the best TPR at the strict FAR threshold. MobileNetV2 follows a similar pattern but drops off more at the threshold level, which shows how model capacity starts to matter more under strict security constraints.

Metric	VGG16 Frozen	CNN	Fine Tuned VGG16	Fine Tuned MobileNetV2
AUC	0.7174	0.50	0.9956	0.9829
EER	33.80%	50.00%	3.11%	6.40%
TPR @ FAR ≤ 0.5%	8.49%	0.00%	84.91%	59.43%
Model Size	N/A	4.63	113.50	16.43

Compared to the original EcoVault targets, none of the models fully meet the required standard. Fine-tuned VGG16 comes closest from a security standpoint, while MobileNetV2 is the more realistic option when considering size and speed, highlighting the tradeoff between performance and deployment.

Robustness, Liveness, and Deployment

To evaluate robustness, I tested performance across bright and dark conditions using the labels created earlier. Fine-tuned VGG16 remained very stable, with AUC values of 0.9936 on bright images and 0.9981 on dark images, along with True Positive Rates of 84.29 percent and 86.11 percent at a False Accept Rate less than or equal to 0.5 percent. MobileNetV2 also stayed consistent in AUC, with values of 0.9843 on bright images and 0.9789 on dark images, but its True Positive Rate at the strict threshold was noticeably lower, ranging from about 54 to 64 percent.

Liveness detection was not implemented in this phase, which is an important limitation for a real system. Even with a PIN, the model could still be vulnerable to spoofing attacks such as printed photos or replayed images. This would need to be addressed before any real deployment. In terms of on-device performance, there were clear tradeoffs between speed and model size. Frozen VGG16 had a latency of 306.1 milliseconds, the CNN ran at 61.4 milliseconds, fine-tuned VGG16 ran at 215.0 milliseconds, and MobileNetV2 ran at 72.9 milliseconds. This made MobileNetV2 the best balance between speed and size, while VGG16 offered stronger verification performance at a higher cost.

Recommendation & Future Steps

Based on these results, the best practical option for EcoVault is fine-tuned MobileNetV2. It is significantly smaller at 16.43 MB, faster in inference, and much more realistic for an offline field application. While it does not match the verification performance of fine-tuned VGG16 at strict thresholds, it is a better overall fit for deployment constraints.

Fine-tuned VGG16 achieved the strongest results and came closest to the target security metrics, but its 113.50 MB size makes it too heavy for a lightweight mobile system. The biggest limitation of this project is the dataset itself. LFW images are low resolution, and even the enrolled user class is relatively small compared to real-world face verification systems. In addition, the models were adapted from general ImageNet pretraining instead of using face-specific architectures like FaceNet or ArcFace.

A stronger next version of this system would use higher resolution images, include more enrollment diversity across sessions, rely on a face-specific pretrained model, and directly evaluate liveness and spoof resistance.